2.4 Stage 2: voxel-level refinement (trim and recall)

Stage 1 decides, per lesion, whether a tumor is present, where it is and which organs may host it; its delineation is a choice among the agents' outlines (Section 2.3, Image grounding). On the evaluated scans both under- and over-segmentation are common even when the location is right: an outline leaks into normal tissue, or covers only part of a large tumor, sometimes a small part of a tumor that has replaced most of its organ. Stage 2 redraws the tumor of every accepted and uncertain lesion at voxel level, inside a region of interest given by stage 1. It removes false-positive voxels (*trim*) and adds false-negative voxels (*recall*). Recall looks for tumor tissue that the segmentation agents missed; it never re-admits the agents' discarded proposals (claims, crumbs or lobes that stage 1 did not keep), which would re-introduce their false positives. Stage 2 does not change any stage-1 decision; it reports how the refinement changes the tumor evidence.

Inputs from stage 1. For a lesion L with decision accepted or uncertain: its anchors and their two-agent voxels, its host credible set $\mathcal{S}_L$ and host probabilities, the territories Θ_h and normal tissue $\mathcal{N}_h$ of the hosts $h \in \mathcal{S}_L$ (the knowledge-base region for an unsegmented host), the chosen extents $\hat{\xi}_a$ of its anchors, $\hat{\xi}_L = \bigcup_{a \in L} \hat{\xi}_a$, and the organ masks with their confidences.

Region of interest. The search region combines a local neighbourhood of the lesion with all of its credible hosts, because the host is often uncertain and a tumor may extend into or arise from any of them:

$$R_L^{(0)} = \Big(\bigcup_{a \in L} a\Big) \oplus r \;\cup\; \bigcup_{h \in \mathcal{S}_L} \tilde{\Theta}_h, \tag{1}$$

restricted to the body. $\tilde{\Theta}_h$ is the territory of h, joined with its knowledge-base region when h is unsegmented or its mask is below the knowledge-base volume range (an organ largely replaced by tumor). The radius r is the 95th percentile, over expert tumors, of the distance from a simulated partial core (below) to the farthest tumor voxel; it is estimated on expert data and not set by hand. A tumor may be larger than any local radius: if the refined tumor touches the boundary of $R_L^{(t)}$ inside the body, the region grows, $R_L^{(t+1)} = R_L^{(t)} \cup (\hat{Z}_L^{(t)} \oplus r)$, and the labelling is repeated until the tumor no longer touches the boundary or reaches the body outline.

Seeds and in-scan appearance models. Earlier tests that compared a candidate with its host failed because the agents select regions that differ from the host whether or not they are tumor. Stage 2 therefore models the tumor from the patient's own confident tumor tissue rather than from the host alone. The *tumor seeds* F_L are the two-agent voxels of L's anchors, eroded by one voxel; a lesion with single-agent anchors only uses their eroded cores and is flagged. The *background seeds* G_L are the normal tissue $\mathcal{N}_h$ of the credible hosts farther than r from the lesion (normal-organ masks may contain unannotated tumor near it), the cores $\hat{H}_o$ of organs outside $\mathcal{S}_L$ with confidence $\mu_o = 1$ inside R_L, and the voxels outside the body. When a host has been replaced by tumor and has no normal tissue, the background comes from the surrounding anatomy only. On each scan, class-conditional densities $p(f \mid T)$ and $p(f \mid B)$ of voxel features $f(v)$ (intensity smoothed in the acquisition plane, and local mean and standard deviation at two scales) are estimated on F_L and G_L. Nothing is learned across patients from agent outputs.

Labelling. Every voxel of R_L receives a tumor probability p_v by a random walker with prior models [Grady, CVPR 2005]: the seeds are fixed ($p_v = 1$ on F_L, 0 on G_L); the unary prior of a free voxel is the posterior from the two densities; the edge weight between neighbours v, w is $\exp\big(-(X_v - X_w)^2/(2\sigma_e^2)\big)$, scaled by λ. The refined tumor is

$$\hat{Z}_L = \mathrm{CC}_{F_L}\big(\{v \in R_L : p_v \geq p^*\}\big) \setminus \bigcup_{o \notin \mathcal{S}_L,\ \mu_o = 1} \hat{H}_o, \tag{2}$$

the component connected to the tumor seeds, without the confident cores of organs outside the credible set; organs inside $\mathcal{S}_L$ are allowed, so that a tumor may extend across its possible hosts. The tumor seeds are never removed. Lesions whose refined tumors touch are merged and the merge is reported.

Trim and recall. With $\hat{\xi}_L$ the stage-1 extent, the trimmed voxels are $\hat{\xi}_L \setminus \hat{Z}_L$ (false positives of the agents' outline) and the recalled voxels are $\hat{Z}_L \setminus \hat{\xi}_L$ (tumor the agents missed). Both are reported per lesion, with their volumes and the host organs they lie in.

Effect on the tumor evidence. Decisions are left unchanged. For every lesion, the existence evidence of stage 1 (Section 2.3, Image grounding) is recomputed on the refined tumor under each host of $\mathcal{S}_L$, with the same expert tables and the same level α, and reported next to the stage-1 outcome together with the mean voxel probability $\bar{p}$ of the stage-1 extent and of the refined tumor. Whether refined evidence should change decisions is decided after this comparison.

Calibration on expert data. The free settings $\theta = (r, \lambda, \sigma_e, p^*)$ and the seed erosion are set on expert data only: R2Seg images with expert tumor masks, organ masks from TotalSegmentator and R2Seg, and simulated agent outputs. Over-segmentation is simulated by dilated and leaking outlines and by outlines merged with the organ; under-segmentation by partial cores of 5–80% of the tumor, including small anchors inside large tumors; a lost host by removing the organ mask where the tumor occupies it. r is the stated percentile; λ, σ_e and p^* maximise the held-out voxel Dice, with collections held out in turn. Results are reported per organ class and modality as voxel precision and recall, Dice and the over- and under-segmented fractions, against the stage-1 extent, the anchor alone and the union of the agents' outlines.

Go/no-go. Stage 1 chose rather than redrew because a host-normalised voxel classifier agreed with experts only at Dice 0.67–0.85 (Section 2.3, Image grounding). Stage 2 differs in using the patient's own tumor tissue and image edges, but the same limits may apply. Refinement is therefore used only for the organ classes and modalities in which it improves the held-out Dice over the stage-1 extent; elsewhere the stage-1 extent is kept and this is reported.